

Class: B.Tech. VI-Semester**Branch:** Computer Science and Engineering**Teaching Scheme :**

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Examination Scheme :

Continuous Internal Evaluation	40 marks
End Semester Examination	60 marks

Course Learning Objectives(LOs):

This course will develop students' knowledge in/on...

LO1: time and space complexity, asymptotic notations, set operations, problem solving with divide and conquer strategy

LO2: greedy and backtracking methods to solve computational problems

LO3: principle of optimality and problem solving with dynamic programming method

LO4: branch and bound method, classes of P, NP, NP-Hard and NP-Complete

UNIT-I (9)

Introduction: Algorithm analysis, Performance analysis, Space complexity and time complexity, Big 'O' notation, Omega notation, Theta notation, Different mathematical approach's for solving time complexity of algorithms

Sets and Disjoint Set Union: Introduction, Union, Find operations

Divide and Conquer: General method, Binary search, Merge sort, Quick sort, Strassen's matrix multiplication

UNIT-II (9)

Greedy Method: General method, Knapsack problem, Job sequencing with deadlines, Optimal storage on tapes, Optimal merge patterns, Single source shortest paths

Back Tracking: General method, N-Queens problem, Sum of subsets, Graph coloring problem

UNIT-III (9)

Dynamic Programming: General method, Multistage graphs, All pairs shortest paths, Single source shortest paths, Optimal binary search trees, String editing, 0/1 Knapsack problem, Reliability design problem, Travelling sales person problem

UNIT-IV (9)

Branch and Bound: General method, Least cost (LC) search, The 15-puzzle problem, Control abstractions for LC search, 0/1 Knapsack problem, Travelling sales person problem

NP Hard and NP Complete Problems: Basic concepts - Nondeterministic algorithms, The classes NP hard and NP complete; COOK's theorem, NP hard graph problems - Clique decision problem, Node cover decision problem, Traveling sales person decision problem

Text Book:

[1] E.Horowitz, S.Sahni, S.Rajasekaran, *Fundamentals of Computer Algorithms*, 2nd ed. Hyderabad: Universities Press, 2018

Reference Books:

- [1] Thomas H.Cormen, Charles E.Leiserson, Ronald L.Rivest, Clifford Stein, *Introduction to Algorithms*, 3rd ed, New Delhi: Prentice-Hall of India, 2010
- [2] Gajendra Sharma, *Design and Analysis of Algorithms*, 4th ed, Khanna, Rajput: Publishing, 2019
- [3] S.Sridhar, *Design and Analysis of Algorithms*, 3rd ed, Oxford University Press, UK: 2015

[4] Mark Allen Weiss, *Data Structures and Algorithm Analysis in Java*, 3rd ed, New Delhi: Pearson, 2012

[5] Rajiv Chopra ,Shipra Raheja, *Design and Analysis of Algorithms*, New Delhi: New Age International Publishers, 2019

Course Learning Outcomes(COs):

On completion of this course, students will be able to...

CO1: *apply divide and conquer strategy for searching and sorting techniques with performance*

CO2: *analyze algorithms using greedy and backtracking methods*

CO3: *design of algorithms using dynamic programming approach*

CO4: *evaluate the classes P, NP ,NP-Hard and NP-Complete*

Course Articulation Matrix (CAM): U18CS604 DESIGN AND ANALYSIS OF ALGORITHMS

CO		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O 2	PSO 3
CO1	U18CS604.1	1	2	2	1	-	1	-	-	1	1	-	1	2	1	2
CO2	U18CS604.2	1	2	2	1	-	1	-	-	1	1	-	1	2	1	2
CO3	U18CS604.3	1	2	2	1	-	1	-	-	1	1	-	1	2	1	2
CO4	U18CS604.4	1	2	2	1	-	1	-	-	1	1	-	1	2	1	2
U18CS604		1	2	2	1	-	1	-	-	1	1	-	1	2	1	2